Sharath Reddy Madhirala

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Gen AI / Data science/Machine Learning Engineer github:https://github.com/sharathreddy203

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SUMMARY:

- **8+ years** of experience in the IT Industry, including 2+years in embedded systems and 6 years specializing in Data Science, Analytics, Generative AI and data-driven solutioning.
- Expertise in Statistics, Machine learning concepts and Data analysis.
- Experience in GenAI, Python Frameworks like Scikit-learn, PySpark, TensorFlow, Keras.
- Expertise in developing and deploying machine learning models, including Linear Regression, Logistic Regression, Ensemble Methods, K-Means Clustering, and KNN, utilizing AWS Cloud Services for model deployment and scaling.
- Hands-on experience in Deep Learning, Natural Language Processing (NLP) and Generative AI (GenAI) using LLM's, GPT, LLaMA, Falcon, BART, T5, RoBERTa applying cutting-edge techniques to solve complex real-world problems.
- Research & Innovation in AI.
- Apply retrieval-augmented generation (RAG) techniques to improve AI performance.
- Expertise in LangChain, LangSmith and LangGraph, CrewAI frameworks for developing advanced GenAI / Agentic AI applications.
- Implement and optimize AI models using Python and frameworks like TensorFlow, LangChain.
- Hands on experience on DB operations like Sql, Nosql and vector DB (Croma, Pinecone, FIASS).
- Experience in creating mature Data science pipelines encompassing Data standardization, Feature extraction, Model validation and optimization.
- Hands on experience on Model Deployment using Kube Flow.
- Hands on experience on C#, Azure cloud.
- Experience on Databricks.
- Hands on experience on AWS services like SageMaker, Glue, Lambda and EC2, Step Function, Bedrock Data Auto. etc.
- Collaborate with stakeholders across the organization to align AI/ML initiatives with Business goals.

TECHNICAL SKILLS:

Programming /Scripting:	Python, C, C++, C#
Tools /IDE	Pycharm, Jupyter Notebook, vs code, Kube flow
Data Science	Machine learning, Statistical modelling, pandas, numpy, scikit-learn, Data visualisation, matplotlib, seaborn, plotly, cuflinks
Operating system	Windows, Linux

EDUCATION

- Pursuing M-tech (Online) : AIML – IIIT-Hyderabad – 2024 to 2026 (Final semester)
- Bachelor of Technology - Auroras Technological & Research Institute - 2012-2015
- Diploma - Govt.Polytechnic – Mahbubnagar - 2009-2012

PROFESSIONAL EXPERIENCE

• Object Win Technologies (Client PwC)	Lead Engineer	[Apr-2025 to Till date]
• Carrier Technologies India Pvt. Ltd	Senior Software Engineer	[Spt-2022 to Dec-2024]
• Votary Softest Sol (Payroll), Qualcomm(Client)	Software Engineer	[Dec-2018 to Aug-2022]
• Eureka IT Solutions Pvt Ltd.	Embedded Software Engineer	[July-2016 to Nov-2018]

PROJECTS:

Project: AI-Powered Document Analysis Pipeline for Pharmaceutical Marketing Compliance

- Developed an AI-driven document analysis and content extraction pipeline to process and analyse pharmaceutical marketing materials such as brochures, emails, and digital campaigns. The system automatically extracts, classifies, and structures textual and visual content from PDFs using a combination of computer vision (YOLO) and cloud-native document AI (AWS Bedrock Data Automation).

(Python, FastAPI, BDA, step function, Lambda, opensearch, S3, RAG, CV (YOLO), GenAI, LLM, Promts, PSQL, DynamoDB, langchain)

Project: AI-Driven Personalized Content Generation Engine for Enterprise B2B Customer Engagement

- Developed an enterprise-scale AI Text Generation Engine for The Coca-Cola Company (TCCC) to deliver hyper-personalized content for B2B customer ordering experiences. Developed a comprehensive GenAI text and Image solutions that ingests multiple data sources including customer profiles, product recommendations, and real-time contextual data to generate personalized messaging, serving multiple bottlers across different geographies with customizable brand guidelines and compliance validation. Developed automated content validation and distribution

pipeline ensuring brand compliance while delivering personalized "why buy this product" messaging to enhance customer purchasing decisions.

- **Backend:** ASP.NET Core, C#
- **Cloud Services:** Azure Blob Storage, Azure Service Bus (implied from messaging patterns)
- **Security:** JWT Authentication, Policy-based Authorization
- **Data Processing:** Large CSV file processing, Streaming uploads, Batch processing
- **Patterns:** Repository pattern, Dependency Injection, Async/Await patterns

CARRIER TECHNOLOGIES INDIA PVT. LTD – [Spt-2022 to Nov-2024]

Project: Excess Energy Consumption Alert

- Developed an excess energy consumption alert system utilizing real-time data from home energy consumption. The system compares the current energy consumption of a home with its past consumption baseline and with the current consumption of neighbouring homes along with their respective historical baselines. This enables timely detection of abnormal energy usage patterns and triggers alerts for potential energy wastage or inefficiencies.

(Python, Pandas, Numpy, Matplotlib, ML Algo (KNN), Statistical Techniques, Snowflake, AWS Services)

Project: Guided Trouble Shooting for Service Technicians

- Developed a Service tech application aimed at efficiently collecting field service data. Implemented advanced analytics to analyse the gathered data, enabling the identification of root causes for failures associated with each fault code encountered during field service operations.

(Python, Analytics, AWS Services)

Project: Chabot for Customer support

- Developed a Chabot system for handling customer queries, improving customer support and user experience. Integrated the Chabot into the organization's mobile app, enabling seamless interaction between customers and the support system. Utilized natural language processing (NLP) and machine learning models to understand and respond to diverse customer inquiries effectively.

(Python, AI, LangChain, RAG, LLM's, AWS Services, Tensorflow, PyTorch, Vector Data Base)

Project: Fault detection in wafers using sensor data

This project is to build a classification model to predict the quality of wafer based on historical sensor data

- Preparing the test Automation scripts & environment setup for the Automation
- Validated & Pre-processed the data
- Segmented the data by creating clusters
- Build & train different Machine Learning models
- Selected the best predicted model based on confusion matrix & AUC score

(Python, Pandas, Numpy, Matplotlib, K-means, Random Forest, XGBoost)

VOTARY SOFTECH SOL (Payroll), QUALCOMM (Client location) – [Dec-2018 to Aug-2022]

Project: Qualcomm Integration and Automation Testing

Embedded Systems software Integration & Automation script development and testing in Qualcomm India pvt ltd through Votay tech as Engineer-1 as part of QCT Product & Tools Software I&T team

- Collaborated with a team to develop machine learning models and implement data visualization techniques, providing actionable insights and supporting data-driven decision-making.
- Worked on Statistics, Machine learning concepts and Data analysis.
- Standardization, Feature extraction, Model validation and optimization.
- Utilized natural language processing (NLP) and machine learning models to understand and respond to diverse customer inquiries effectively.
- Having experience in bring up the multiple variant targets in Qualcomm such as SDX55, SDX50Mv2, SDM632/439/29, MSM8998/96 and 6k targets

(Python, Pandas, Numpy, Matplotlib, Seaborn, ML algo's,)

EUREKA IT SOLUTIONS PVT LTD. – [July-2016 to Nov-2018]

Project: IOT Solutions

This project is to define a framework for collecting/sending the sensor data at regular intervals to cloud, it then processes the data at back end and generate different kinds of reports & indicate to customer/user in their web/mobile dashboard.